

A GAP-TWO INEQUALITY FOR LINEAR EXTENSIONS AND A POLYNOMIAL BOUND FOR THE BALANCE CONSTANT OF POSETS OF BOUNDED RANGE

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ABSTRACT. Let f be a uniformly random linear extension of a finite poset P , and let $x \parallel y$. We show that $\mathbb{P}[f(x) - f(y) \geq 2] \geq \mathbb{P}[f(y) < f(x)]/(\pi_N + 1)$ whenever some element can lie between y and x . Here π_N is the largest number of elements incomparable to such an element. The constant is sharp for every value of π_N . Together with a linear lower bound on the displacement and a linear replacement of Lemma 4.2 of Aires, Chan, Pak and Panova [KSBFT], this removes every exponentially small quantity from the proof of their Theorem 1.3. The result: every finite non-chain poset P of range $\pi(P) \leq D$ satisfies

$$\delta(P) \geq C_{\text{BFT}} + \min\left(\theta_0, \frac{C_{\text{BFT}}}{(5 + 3\sqrt{5})(D + 1) + 1}\right), \quad C_{\text{BFT}} = \frac{5 - \sqrt{5}}{10}, \quad \theta_0 = 0.2764 - C_{\text{BFT}} \approx 6.80 \cdot 10^{-6}.$$

This replaces the bound $C_{\text{BFT}} + (D + 1)^{-4(D+1)}/4096$ of [KSBFT]. In particular $\delta(P) > 0.2764$ for every $D \leq 3471$. For $D \leq 6$ the stronger bound $\delta(P) \geq 1/3$ is already known [BW92, Pec08], so the bound is new only for $D \geq 7$.

1. STATEMENTS

Throughout, $P = (X, \prec)$ is a finite poset with $n = |X|$, and f is a uniformly random linear extension, viewed as a bijection $X \rightarrow [n]$. We write $u \parallel v$ when u, v are incomparable, $\pi(u) = \#\{v : v \parallel u\}$, and $\pi(P) = \max_u \pi(u)$ (the *range*). We set $h(u) = \mathbb{E}f(u)$, $\delta(P; u, v) = \min(\mathbb{P}[f(u) < f(v)], \mathbb{P}[f(v) < f(u)])$, and $\delta(P) = \max_{u \neq v} \delta(P; u, v)$.

Definition 1.1. For $x \parallel y$ put $N(x, y) = \{w \in X \setminus \{x, y\} : w \not\prec y \text{ and } x \not\prec w\}$ and $\pi_N = \max_{w \in N(x, y)} \pi(w)$.

Theorem 1.2 (Lemma W). *Let $x \parallel y$ in a finite poset.*

- (a) *Some linear extension g has $g(y) < g(w) < g(x)$ if and only if $w \in N(x, y)$. Hence $\mathbb{P}[f(x) - f(y) \geq 2] > 0$ if and only if $N(x, y) \neq \emptyset$.*
- (b) *If $N(x, y) \neq \emptyset$, then $\#\{g : g(x) = g(y) + 1\} \leq \pi_N \cdot \#\{g : g(x) = g(y) + 2\}$, and consequently*

$$\mathbb{P}[f(x) - f(y) \geq 2] \geq \frac{\mathbb{P}[f(y) < f(x)]}{\pi_N + 1} \geq \frac{\mathbb{P}[f(y) < f(x)]}{\pi(P) + 1}.$$

- (c) *The constant is sharp: there are posets with $\pi_N = 1$ and equality in (b), and for every $D \geq 2$ there are posets with $\pi_N = \pi(P) = D$ and equality in (b). For $D \geq 3$ this holds even when (x, y, z) is a Case D BFT triple (Example 2.2).*

Theorem 1.3 (Theorem 1.3''). *Let $D \geq 2$ and let P be a finite poset that is not a chain, with $\pi(P) \leq D$. Put*

$$\theta_0 := 0.2764 - C_{\text{BFT}} = 6.79775 \dots \cdot 10^{-6}, \quad \theta_W(D) := \frac{C_{\text{BFT}}}{(5 + 3\sqrt{5})(D + 1) + 1}.$$

If $0 \leq \theta \leq \theta_0$ and $\theta < \theta_W(D)$, then $\delta(P) > C_{\text{BFT}} + \theta$. Consequently

$$\delta(P) \geq C_{\text{BFT}} + \min(\theta_0, \theta_W(D)),$$

with strict inequality when $\theta_W(D) > \theta_0$, which happens exactly for $D \leq D_0 = 3471$. For those D the bound reads $\delta(P) > 0.2764$. For $D > 3471$ it reads $\delta(P) \geq C_{\text{BFT}} + \theta_W(D) \approx C_{\text{BFT}} + 0.02361/(D + 1)$.

For comparison, [KSBFT, Thm. 1.3] gives $\delta(P) > C_{\text{BFT}} + (D + 1)^{-4(D+1)}/4096$. For $D \leq 6$ the stronger bound $\delta(P) \geq \frac{1}{3}$ is known: range ≤ 5 by Brightwell and Wright [BW92] and range ≤ 6 by Peczarski [Pec08] (as recorded in [KSBFT, p. 3]; we have not read either paper). So Theorem 1.3 is new only for $D \geq 7$.

What is imported. Theorem 1.2 is self-contained. Theorem 1.3 uses exactly the following four results from the literature, and nothing else. We state them in the form used. A BFT triple is a triple (x, y, z) of distinct elements, not forming a chain, with $h(x) \leq h(y) \leq h(z) \leq h(x) + 2$. Since $u \prec v$ implies $h(u) < h(v)$, the only possible comparabilities are $x \prec y$, $x \prec z$ and $y \prec z$. Up to duality and reversal of the triple, one of the following holds [BFT95, §4]: Case A ($x \prec z$, $y \prec z$, $x \parallel y$); Case B ($y \prec z$, $x \parallel y$, $x \parallel z$); Case C (pairwise incomparable); Case D ($x \prec z$, $x \parallel y$, $y \parallel z$). Here “duality” means passing to the dual poset P^* and replacing (x, y, z) by (z, y, x) . This preserves δ , π and the defining conditions of a BFT triple (since $h_{P^*}(u) = n + 1 - h_P(u)$), so any statement proved for one of the four cases applies equally to its dual.

Imported fact I1 (Kahn–Linial [KL91, p. 365]; general form [AK25a, Cor. 5.1(b)]; [KSBFT, Lemma 2.1]). *If $x \neq y$ and $h(x) \leq h(y)$, then $\mathbb{P}[f(y) < f(x)] \leq 1 - e^{-1}$.*

Imported fact I2 ([BFT95, Thms. 4.1, 4.2]). *In Cases A and B, $\mathbb{P}[f(y) < f(x)] \geq 1/3$.*

Imported fact I3 ([KSBFT, L. 2.3, App. A]; [BFT95, §6]). *In Case C, $\max(\delta(P; x, y), \delta(P; y, z)) > 0.2764$.*

Imported fact I4 ([BFT95, (5.2), (5.5), (5.7), (5.8)], as [KSBFT, (2.1)–(2.2)]). *In Case D, put $p(i, j) = \mathbb{P}[f(y) - f(x) = i, f(z) - f(y) = j]$, $p = \mathbb{P}[f(y) < f(x)]$, $p' = \mathbb{P}[f(z) < f(y)]$, $S = p + p'$, $a_1 = p(1, 1)$, $a_2 = p(1, 2) + p(2, 1)$, $a_3 = p(2, 2)$. Then*

$$2a_1 + a_2 \leq S, \quad 2a_2 + 2a_3 \leq S, \quad 4a_1a_3 \leq a_2^2, \quad 7 \leq 9S + 7a_1 + 3a_2 + 3a_3.$$

Everything else from [KSBFT] that we use is reproved below: Cor. 2.2, Lemma 2.4, Lemma 3.2 (with a weaker hypothesis and a new proof of (iii); we omit its part (iv)), Lemma 4.1, and the identities (4.2)–(4.7). Neither Lemma 3.1 (the $q_m = (D+1)^{-m(D+1)}$ bound) nor Lemma 4.2 (the $M^2/441$ conversion) of [KSBFT] is used. Lemma 4.2 is replaced by a linear conversion (Proposition 5.3), and the lower bound (4.11) on M by Proposition 5.2. Section 7 records our check of the proof of Fact I3.

2. PROOF OF LEMMA W

Proof of Theorem 1.2(a). If $g(y) < g(w) < g(x)$ then $w \not\prec y$ and $x \not\prec w$, so $w \in N(x, y)$. Conversely, let $w \in N(x, y)$. Adding the relations $y < w < x$ to $\prec$ creates a cycle only if $w \preceq y$, $x \preceq w$ or $x \preceq y$, and none of these holds. So the transitive closure is a partial order, and any of its linear extensions will do. Finally, $f(x) - f(y) \geq 2$ means some element sits strictly between y and x , and that element lies in $N(x, y)$. $\square$

Proof of Theorem 1.2(b). Let $E_k = \{g : g(x) = g(y) + k\}$ and $N = N(x, y) \neq \emptyset$. We define $\Phi : E_1 \rightarrow E_2$. Fix $g \in E_1$, with y at position i and x at position $i + 1$. Every element of N lies before y or after x . Put $A = \{w \in N : g(w) > g(x)\}$.

Case $A \neq \emptyset$. Let w be the first element of A in g , say $g(w) = i + 1 + j$ with $j \geq 1$. Let $t_0 = x, t_1, \dots, t_{j-1}$ be the elements in positions $i + 1, \dots, i + j$. We claim that each t_k is incomparable to w . First, $w \not\prec t_k$ because t_k precedes w in g . Suppose $t_k \prec w$. Then $k \neq 0$, because $x \not\prec w$. Also $t_k \not\prec y$, because t_k comes after y in g ; and $x \not\prec t_k$, since otherwise $x \prec t_k \prec w$. Hence $t_k \in N$ and t_k lies after x and before w , which contradicts the choice of w . Moving w to position $i + 1$ changes the relative order only of the incomparable pairs $\{w, t_k\}$. The result is therefore a linear extension, with y, w, x in positions $i, i + 1, i + 2$. Let $\Phi(g)$ be this extension.

Case $A = \emptyset$. Then N lies entirely before y . Let w be the last element of N in g , and let $s_1, \dots, s_{j-1}, s_j = y$ be the elements after w up to and including y . Again each s_k is incomparable to w . First, $s_k \not\prec w$ by position. Next, $w \not\prec y$ because $w \in N$. Finally, suppose $w \prec s_k$ with $k < j$. Then $s_k \not\prec y$, since otherwise $w \prec y$; $x \not\prec s_k$, because s_k precedes x ; and $s_k \neq x, y$. So $s_k \in N$ lies after w , which contradicts maximality. Moving w to just after y gives a linear extension in which y, w, x are consecutive. Let $\Phi(g)$ be this extension. The case split matters: if we moved the last N -element before y even when $A \neq \emptyset$, then Φ would not be a function of g .

Multiplicity. Fix $g' \in E_2$ and let w be its middle element, so $w \in N$. Any Φ -preimage of g' is obtained from g' by moving w in one of two ways:

- right by j places, past x and $j - 1$ further elements (the case $A \neq \emptyset$);
- left by j places, past y and $j - 1$ further elements (the case $A = \emptyset$).

So a preimage is determined by the side and by j . We showed above that in a preimage all j elements that w passes are incomparable to w . Let a_R and a_L be the lengths of the maximal runs of elements incomparable

to w immediately to the right and to the left of w in g' . Then $j \leq a_R$ on the right side and $j \leq a_L$ on the left side. If $y \prec w$, then $a_L = 0$, which only helps. The two runs are disjoint sets of elements incomparable to w , so

$$|\Phi^{-1}(g')| \leq a_R + a_L \leq \pi(w) \leq \pi_N.$$

Hence $|E_1| \leq \pi_N |E_2|$, and

$$\mathbb{P}[f(y) < f(x)] = \mathbb{P}[E_1] + \mathbb{P}[f(x) - f(y) \geq 2] \leq \pi_N \mathbb{P}[E_2] + \mathbb{P}[f(x) - f(y) \geq 2] \leq (\pi_N + 1) \mathbb{P}[f(x) - f(y) \geq 2].$$

Note that $\pi_N \geq 1$: an element $w \in N$ with $y \prec w \prec x$ would force $y \prec x$. $\square$

Remark 2.1. The proof gives $\mathbb{P}[f(y) < f(x)] \leq \pi_N \mathbb{P}[f(x) = f(y) + 2] + \mathbb{P}[f(x) - f(y) \geq 2]$, not $\mathbb{P}[f(x) = f(y) + 2] \geq \mathbb{P}[f(y) < f(x)]/(\pi_N + 1)$, and the latter is false in general. Take a chain $c_1 \prec c_2 \prec x$ and an element y incomparable to all three. Then $N(x, y) = \{c_1, c_2\}$, $\pi_N = 1$, $\mathbb{P}[f(y) < f(x)] = 3/4$ and $\mathbb{P}[f(x) = f(y) + 2] = 1/4 < 3/8$, while $\mathbb{P}[f(x) - f(y) \geq 2] = 1/2 \geq 3/8$. Only the bound on $\mathbb{P}[f(x) - f(y) \geq 2]$ is used below.

Example 2.2 (sharpness, Theorem 1.2(c)). *The case $\pi_N = 1$.* Take $y \prec w$ and an element x incomparable to both. Then $N(x, y) = \{w\}$ and $\pi(w) = 1$. Of the three linear extensions, two have y before x and one has y, w, x in this order, so the ratio in (b) is $\frac{1}{2} = 1/(\pi_N + 1)$.

Generic family, $D \geq 2$. Take a chain $x = c_1 \prec \dots \prec c_{D-1}$ and two further elements y, w incomparable to everything. Then $\pi(y) = \pi(w) = D = \pi(P)$ and $N(x, y) = \{w\}$. The event $f(y) < f(x)$ says that y is first among $\{y\} \cup \{c_i\}$, and then w can occupy any of $D + 1$ slots. Exactly one of these slots lies between y and x . So the ratio in (b) is $1/(D + 1)$.

Case D BFT triple, $D \geq 3$. Let Q_k be the $2+1$ poset $\{x \prec z\} \sqcup \{y\}$ with a chain $c_1 \prec \dots \prec c_k$ placed above all of it. Let $P_k = Q_k \sqcup \{w\}$ with w incomparable to everything, and put $D = k + 3$. Then $\pi(P_k) = \pi(w) = D$. In Q_k the heights are $(h(x), h(y), h(z)) = (4/3, 2, 8/3)$. Inserting w uniformly into the $D + 1$ slots multiplies every height by $(D + 2)/(D + 1) \leq 5/4$. So (x, y, z) is a Case D BFT triple: $h(z) - h(x) \leq 5/3$. Moreover $N(x, y) = \{w\}$, since z and every c_i lie above x . In Q_k , y before x forces y immediately before x . Hence $\mathbb{P}[f(y) < f(x)] = 1/3$ and $\mathbb{P}[f(x) - f(y) \geq 2] = 1/(3(D + 1))$, and the ratio is $1/(D + 1)$.

Remark 2.3 (relation to known inequalities). Let $a_i = \mathbb{P}[f(y) - f(x) = i]$ and $b_i = \mathbb{P}[f(x) - f(y) = i]$. Kahn and Saks [KS84] (as restated in [BFT95, Lemma 2.1]) prove $a_1 = b_1$ and the two-sided inequality $a_2 + b_2 \geq a_1 + b_1$. Lemma W is one-sided: $b_1 \leq \pi_N b_2$. It carries a range factor, which Example 2.2 shows cannot be replaced by an absolute constant. In the notation of [BFT95] it says $\epsilon = b_1/B \leq \pi_N/(\pi_N + 1)$, where $B = \sum_i b_i$. We have not found Lemma W in the literature, but our search was limited. We searched the full text of the Chan–Pak survey [CP23], of recent work of Chan, Pak and Panova, of van Handel, Yan and Zeng [vHYZ23], and of [AK25a, AK25b, Haq26], as well as the statements of [BFT95]. We did *not* read Kahn–Saks [KS84] (only its restatement in [BFT95]), Kahn–Linial [KL91], Brightwell–Wright [BW92], Peczarski [Pec08] or Brightwell’s survey [Bri99]; these were not accessible to us. A statement of this kind in one of them is not ruled out.

Remark 2.4. Lemma W bounds a *ratio*. Absolute probabilities of such events can be exponentially small in the range. For example, for two parallel k -chains the probability that the top of one precedes the bottom of the other is $\binom{2k}{k}^{-1}$. This is why [KSBFT, Lemma 3.1], which bounds absolute probabilities, loses $(D + 1)^{-m(D + 1)}$. In the application below the denominator $\mathbb{P}[f(y) < f(x)]$ is bounded below by a constant (Lemma 3.3), so the ratio bound is all that is needed.

3. CASE D NEAR C_{BFT}

Lemma 3.1 ([KSBFT, Cor. 2.2]). *If $\delta(P) \leq 0.2764$ and $h(x) \leq h(y)$, then $\mathbb{P}[f(y) < f(x)] \leq 0.2764$.*

Proof. If $x \prec y$ the probability is 0. Otherwise suppose $\mathbb{P}[f(y) < f(x)] > 0.2764$. By Fact I1, $\mathbb{P}[f(x) < f(y)] \geq e^{-1} > 0.2764$. Then $\delta(P; x, y) > 0.2764$, which is a contradiction. $\square$

Lemma 3.2 ([KSBFT, Lemma 2.4]). *In Case D, with the notation of Fact I4 and $R := S - 2a_1 - a_2$ (≥ 0):*

- (a) $S/2 \geq C_{\text{BFT}} + R/(5 + 3\sqrt{5})$. *This part needs no hypothesis on $\delta(P)$.*
- (b) $R \geq \mathbb{P}[f(x) - f(y) \geq 2] + \mathbb{P}[f(y) - f(z) \geq 2]$.
- (c) *If $\delta(P) \leq 0.2764$, then $\delta(P; x, y) = p$, $\delta(P; y, z) = p'$, and $\max(p, p') \geq C_{\text{BFT}} + R/(5 + 3\sqrt{5})$.*

Proof. (a) Since $S > 0$, put $b_i = a_i/S$ and $s = 1 - R/S = 2b_1 + b_2 \in [0, 1]$. Fact I4 gives $b_2 + b_3 \leq \frac{1}{2}$ and $4b_1b_3 \leq b_2^2$. Let $F(s) = \max(\frac{7}{2}s, \frac{7}{2}\sqrt{s^2 + 1/4} - \frac{1}{4})$. We claim $7b_1 + 3b_2 + 3b_3 \leq F(s)$. If $b_1 = 0$, the left side is $3(b_2 + b_3) \leq 3/2 \leq F(s)$. If $u := b_1 > 0$, then $u \leq s/2$ and $b_2 + b_3 \leq b_2 + b_2^2/(4u) = s^2/(4u) - u$, so the left side is at most $7u + 3 \min(\frac{1}{2}, s^2/(4u) - u)$. The two terms in the minimum are equal at $u_0 = (\sqrt{1 + 4s^2} - 1)/4 \leq s/2$. On $(0, u_0]$ the bound is $7u + \frac{3}{2}$, which is increasing, with value $\frac{7}{2}\sqrt{s^2 + 1/4} - \frac{1}{4}$ at u_0 . On $[u_0, s/2]$ it is $4u + 3s^2/(4u)$, which is convex, so its maximum is at an endpoint; the value at $s/2$ is $\frac{7}{2}s$. Since F is convex with $F(0) = \frac{3}{2}$ and $F(1) = c := (7\sqrt{5} - 1)/4$, we get $F(s) \leq c - (c - \frac{3}{2})(1 - s)$. Hence

$$7 \leq 9S + S(7b_1 + 3b_2 + 3b_3) \leq S(9 + F(s)) \leq (9 + c)S - (c - \frac{3}{2})R.$$

The claim follows from $7/(2(9 + c)) = C_{\text{BFT}}$ and $(c - \frac{3}{2})/(2(9 + c)) = 1/(5 + 3\sqrt{5})$.

(b) Swapping adjacent incomparable elements gives $p(-1, j) = p(1, j - 1)$ and $p(j, -1) = p(j - 1, 1)$ for $j \geq 2$. Hence $2a_1 + a_2 = p(-1, 2) + p(-1, 3) + p(2, -1) + p(3, -1)$. Since $x \prec z$, the event $\{f(x) = f(y) + 1\}$ forces $f(z) - f(y) \geq 2$, so $\{f(x) = f(y) + 1\} = \bigsqcup_{j \geq 2} \{f(y) - f(x) = -1, f(z) - f(y) = j\}$. Therefore $p - \mathbb{P}[f(x) - f(y) \geq 2] \geq p(-1, 2) + p(-1, 3)$. Symmetrically, $p' - \mathbb{P}[f(y) - f(z) \geq 2] \geq p(2, -1) + p(3, -1)$. Adding the two gives (b).

(c) By Lemma 3.1, $p, p' \leq 0.2764 < \frac{1}{2}$. So $\delta(P; x, y) = p$, $\delta(P; y, z) = p'$, and $\max(p, p') \geq S/2$. Now apply (a). $\square$

Lemma 3.3 (the window forces balance). *In Case D, if $\delta(P) \leq C_{\text{BFT}} + \theta$ with $0 \leq \theta \leq \theta_0$, then $p \geq C_{\text{BFT}} - \theta$ and $p' \geq C_{\text{BFT}} - \theta$.*

Proof. By Lemma 3.2(a), $S \geq 2C_{\text{BFT}}$. By Lemma 3.2(c), $p' \leq \delta(P) \leq C_{\text{BFT}} + \theta$. Hence $p \geq 2C_{\text{BFT}} - p' \geq C_{\text{BFT}} - \theta$, and symmetrically for p' . $\square$

4. STRUCTURE OF BFT TRIPLES IN THE WINDOW

Lemma 4.1 ([KSFT, Lemma 3.2] under a weaker hypothesis). *Let $\pi(P) \leq D$ and $\delta(P) \leq C_{\text{BFT}} + \theta$, where $0 \leq \theta \leq \theta_0$ and $\theta < \theta_W(D)$. Let (x, y, z) be any BFT triple. Then: (i) it is in Case D; (ii) z covers x ; (iii) y is incomparable only to x and z ; (iv) with $L = \{w : w \prec y\}$ and $U = \{w : y \prec w\}$, we have $X = L \sqcup \{x, y, z\} \sqcup U$, and $L \prec z$, $x \prec U$, $L \prec U$.*

Part (iv) here is part (v) of [KSFT, Lemma 3.2]. Their part (iv) is not needed and is omitted.

Proof. (i) We have $\delta(P) \leq C_{\text{BFT}} + \theta_0 = 0.2764$. Cases A and B, and their duals, give a pair with $\delta \geq 1/3$, by Facts I2 and I1 (note $1 - e^{-1} < 2/3$); for the duals, apply Fact I2 in P^* . Case C gives a pair with $\delta > 0.2764$ by Fact I3. Both contradict $\delta(P) \leq 0.2764$.

(ii) Suppose $x \prec w \prec z$. Then $g(z) - g(x) \geq 2$ for every g . The event $f(x) < f(y) < f(z)$ has positive probability, since adding $x < y < z$ creates no cycle. On that event $g(z) - g(x) \geq 3$. So $h(z) - h(x) > 2$, which is a contradiction.

(iii) By Lemma 3.2(c) and the hypothesis, $R \leq (5 + 3\sqrt{5})\theta$. Suppose some g has $g(y) < g(w) < g(x)$. Then $N(x, y) \neq \emptyset$ by Theorem 1.2(a). Theorem 1.2(b), Lemma 3.3 and Lemma 3.2(b) then give

$$\frac{C_{\text{BFT}} - \theta}{D + 1} \leq \frac{p}{\pi_N + 1} \leq \mathbb{P}[f(x) - f(y) \geq 2] \leq R \leq (5 + 3\sqrt{5})\theta.$$

This is equivalent to $\theta \geq \theta_W(D)$, a contradiction. The same argument applies to the ordered pair (y, z) : use Theorem 1.2 with (x, y) replaced by (y, z) , whose denominator is $p' = \mathbb{P}[f(z) < f(y)] \geq C_{\text{BFT}} - \theta$, together with the second summand of Lemma 3.2(b). Hence every g satisfies

$$(1) \quad g(y) < g(x) \Rightarrow g(x) = g(y) + 1, \quad g(z) < g(y) \Rightarrow g(y) = g(z) + 1.$$

By Theorem 1.2(a), (1) says $N(x, y) = N(y, z) = \emptyset$. In other words, every $w \neq x, y$ has $w \prec y$ or $x \prec w$, and every $w \neq y, z$ has $w \prec z$ or $y \prec w$. If some $w \notin \{x, y, z\}$ were incomparable to y , this would give $x \prec w \prec z$, contradicting (ii).

(iv) The partition follows from (iii). For $w \in L$, since $w \not\prec y$, we get $w \prec z$. For $w \in U$, $x \prec w$. Finally, $L \prec U$ holds through y . $\square$

Remark 4.2. In [KSFT, Lemma 3.2(iii)] the chain is $q_3 \leq \mathbb{P}[E] \leq \mathbb{P}[f(x) - f(y) \geq 2] \leq R$, with $E = \{f(y) < f(w) < f(x)\}$ and $q_3 = (D + 1)^{-3(D+1)}$ from their Lemma 3.1. The proof above bounds the *same* middle term, for the same ordered pair, in the same orientation and for the same event as in Lemma 3.2(b), and it

does so directly. So q_3 and $\mathbb{P}[E]$ disappear. This removes the first of the two places where the exponential bound of [KSBFT, Lemma 3.1] enters their proof. The second is their (4.11), $M \geq q_2$, which is replaced by Proposition 5.2; and the square in their Lemma 4.2 is replaced by Proposition 5.3. With Lemma W alone, their argument would still give an improvement of order q_2^2 .

5. THE DISPLACEMENT ARGUMENT WITH LINEAR LOSSES

Let $v_1, \dots, v_n$ list X in non-decreasing order of h , let $d_i = h(v_i) - i$, and let $M = \max_i |d_i|$.

Lemma 5.1 ([KSBFT, Lemma 4.1]). *Suppose $G(P)$, the incomparability graph, is connected and $n \geq 3$. Then some k has (v_k, v_{k+1}, v_{k+2}) a BFT triple with $\max(|d_k|, |d_{k+2}|) = M$.*

Proof. Positions are distinct integers, so $d_1 \geq 0$, $d_1 + d_2 \geq 0$, $d_n \leq 0$ and $d_{n-1} + d_n \leq 0$. Hence either $d_i = M$ for some $i \leq n-2$, or $d_j = -M$ for some $j \geq 3$. Take $k = i$ in the first case and $k = j-2$ in the second. Then $h(v_{k+2}) - h(v_k) = 2 + d_{k+2} - d_k \leq 2$. Suppose the triple were a chain $x \prec y \prec z$. Then $g(z) - g(x) \geq 2$ for every g , and since $h(z) - h(x) \leq 2$, in fact $g(z) - g(x) = 2$ for every g . Any w incomparable to one of x, y, z satisfies $w \not\prec x$ and $z \not\prec w$, so some g has $g(x) < g(w) < g(z)$, which gives $g(z) - g(x) \geq 3$. That is a contradiction. So the triple would be an isolated set in $G(P)$, which contradicts connectedness. $\square$

Proposition 5.2 (displacement floor). *If $G(P)$ is connected, $n \geq 2$ and $\pi(P) \leq D$, then $M \geq d_1 \geq 1/(D+1)$. The bound on d_1 is sharp for a D -chain plus one isolated point. (For that poset and $D \geq 3$, M itself is between $\frac{3}{7}$ and $\frac{1}{2}$.)*

Proof. The element $a = v_1$ is minimal, since $w \prec a$ would give $h(w) < h(a)$. Because $G(P)$ is connected, a has an incomparable element. For g with $g(a) = 1$, let c be the first element incomparable to a , at position $t \geq 2$. Each element u in positions $2, \dots, t-1$ satisfies $a \prec u$. Then $u \not\prec c$, since otherwise $a \prec c$; and $c \not\prec u$ by position. So $u \parallel c$. Moving c to the front therefore gives an extension with c, a in positions $1, 2$. The original g is recovered from the image and from t . Since c passed $t-1$ elements incomparable to it, $t-1 \leq \pi(c) \leq D$. Hence $\#\{g(a) = 1\} \leq D \#\{g(a) = 2\}$. This gives $\mathbb{P}[f(a) = 1] \leq D/(D+1)$ and $d_1 = \mathbb{E}f(a) - 1 \geq \mathbb{P}[f(a) \geq 2] \geq 1/(D+1)$. $\square$

Proposition 5.3 (replaces [KSBFT, Lemma 4.2]). *Assume the conclusions of Lemma 4.1 for the triple $(x, y, z) = (v_k, v_{k+1}, v_{k+2})$ of Lemma 5.1. Then*

$$a := \max(\delta(P; x, y), \delta(P; y, z)) \geq C_{\text{BFT}} + \frac{M}{20 + 2\sqrt{5}}.$$

Proof. Height identities (after [KSBFT, (4.2)–(4.7)]). Let $L' = L \cup \{x\}$ and $U' = U \cup \{z\}$. Then $L' \prec U'$. Every $w \in L$ has $h(w) < h(y)$ and $w \neq x$, so it precedes x in the h -order; symmetrically for U . Hence $x = v_m$ with $m = |L'|$ (so $k = m$). In every linear extension, L' occupies the first m positions (except that y may interleave) and U' the last. The element y sits between the two blocks, or immediately before x when x is last in the L' -block, or immediately after z when z is first in the U' -block.

Let $f_{L'}$ and $f_{U'}$ be independent uniform extensions of $P[L']$ and $P[U']$. Put $r = \mathbb{P}[f_{L'}(x) = m]$, $s = \mathbb{P}[f_{U'}(z) = 1]$, $v = r + s$ and $Z = 1 + v$. Then $e(P) = e(L')e(U')Z$, so $p = r/Z$ and $p' = s/Z$, both at most $\frac{1}{2}$. Hence $a = \max(r, s)/Z$.

Let α be the expected number of elements of L after x , and β the expected number of elements of U before z . Counting the positions of y gives $\alpha = (1 + s)t/Z$ with $t = \mathbb{E}[m - f_{L'}(x)] \geq 1 - r$, and similarly $\beta = (1 + r)u/Z$ with $u \geq 1 - s$. Counting the elements below x and z gives $h(x) = m + p - \alpha$ and $h(z) = m + 2 + \beta - p'$. Hence $d_m = p - \alpha$ and $d_{m+2} = \beta - p'$. The condition $h(z) - h(x) \leq 2$ gives $\alpha + \beta \leq v/Z$. Put $\alpha_* = \alpha - (1 + s)(1 - r)/Z \geq 0$ and $\beta_* = \beta - (1 + r)(1 - s)/Z \geq 0$. Then

$$0 \leq \alpha_* + \beta_* \leq \gamma/Z, \quad \gamma := v + 2rs - 2 \geq 0.$$

Linear conversion. Let $\rho = (\sqrt{5} - 1)/2$, so $\rho^2 = 1 - \rho$, $1 + 2\rho = \sqrt{5}$ and $C_{\text{BFT}} = \rho/\sqrt{5}$. Write $r = \rho + X$ and $s = \rho + Y$. Since $r, s \in [0, 1]$, we have $|X|, |Y| \leq \rho$. Put $W = |X| + |Y|$ and $\varepsilon = a - C_{\text{BFT}}$.

- (1) $r/Z - C_{\text{BFT}} = A/(\sqrt{5}Z)$ with $A = (1 + \rho)X - \rho Y$. Symmetrically $s/Z - C_{\text{BFT}} = B/(\sqrt{5}Z)$ with $B = (1 + \rho)Y - \rho X$. So $\sqrt{5}Z\varepsilon = \max(A, B)$, with $A + B = X + Y$ and $A - B = \sqrt{5}(X - Y)$.
- (2) $\gamma = \sqrt{5}(X + Y) + 2XY$, using $2\rho + 2\rho^2 = 2$. We show $X + Y \geq 0$. Suppose $X + Y < 0$. If $XY \leq 0$, then $\gamma < 0$. If $XY > 0$, both are negative and $2|X||Y| \leq 2\rho|Y| < \sqrt{5}W$, so again $\gamma < 0$. Either way this contradicts $\gamma \geq 0$.

(3) $\max(A, B) = \frac{X+Y}{2} + \frac{\sqrt{5}|X-Y|}{2} \geq \max(X, Y) \geq W/2$, where the last step uses $X + Y \geq 0$. Hence $W \leq 2\sqrt{5}Z\varepsilon$, and in particular $\varepsilon \geq 0$.

(4) We have $d_m = F/Z - \alpha_*$ and $d_{m+2} = G/Z + \beta_*$, where

- $F = r - (1+s)(1-r) = (2+\rho)X - (1-\rho)Y + XY$;
- $G = (1+r)(1-s) - s = (1-\rho)X - (2+\rho)Y - XY$.

Using $|XY| \leq \rho \min(|X|, |Y|)$, both satisfy $|F|, |G| \leq (2+\rho)W$. Also $2|X||Y| \leq \rho W$ gives $\gamma \leq (\sqrt{5} + \rho)W$. Hence $|d_m|, |d_{m+2}| \leq (2+2\rho + \sqrt{5})W/Z = (1+2\sqrt{5})W/Z \leq (20+2\sqrt{5})\varepsilon$.

By the choice of the triple, $M = \max(|d_m|, |d_{m+2}|) \leq (20+2\sqrt{5})\varepsilon$. $\square$

6. PROOF OF THEOREM 1.3

The components of $G(P)$ are totally ordered by $\prec$. Indeed, if $a \prec b$ lie in different components and $a' \parallel a$, then a' is comparable to b , and $b \prec a'$ would give $a \prec a'$; so $a' \prec b$, and connectedness propagates this. So each component K is an interval of every linear extension, and probabilities of events inside K are the same in P and in $P[K]$. Hence $\delta(P) \geq \delta(P[K])$, and $\pi(P[K]) \leq D$. Since P is not a chain, some K has $|K| \geq 2$. If $|K| = 2$, then $\delta(P) \geq \frac{1}{2}$. So we may assume that $G(P)$ is connected and $n \geq 3$.

Suppose $\delta(P) \leq C_{\text{BFT}} + \theta$, with $0 \leq \theta \leq \theta_0$ and $\theta < \theta_W(D)$. Lemma 5.1 gives a BFT triple with $\max(|d_k|, |d_{k+2}|) = M$. Lemma 4.1 applies to it, and Propositions 5.3 and 5.2 give

$$\delta(P) \geq a \geq C_{\text{BFT}} + \frac{M}{20+2\sqrt{5}} \geq C_{\text{BFT}} + \frac{1}{(20+2\sqrt{5})(D+1)}.$$

But $\theta < \theta_W(D) < \frac{C_{\text{BFT}}}{(5+3\sqrt{5})(D+1)} = \frac{0.02360\dots}{D+1} < \frac{0.04086\dots}{D+1} = \frac{1}{(20+2\sqrt{5})(D+1)}$, which contradicts $\delta(P) \leq C_{\text{BFT}} + \theta$. If $\theta_W(D) > \theta_0$, take $\theta = \theta_0$. Otherwise let $\theta \uparrow \theta_W(D)$. $\square$

Numbers (computed in double precision):

- $\theta_0 = 6.797750 \cdot 10^{-6}$;
- $\theta_W(3471) = 6.799026 \cdot 10^{-6} > \theta_0 > \theta_W(3472) = 6.797068 \cdot 10^{-6}$, so $D_0 = 3471$;
- at $D = 10^4$ the gain is $2.36 \cdot 10^{-6}$, and at $D = 10^6$ it is $2.36 \cdot 10^{-8}$.

Remark 6.1 (which constraint binds). Three constraints appear in the proof:

- the cap θ_0 , which comes only from Case C (Fact I3) via Lemma 4.1(i) and Lemma 3.1;
- Lemma W's $\theta_W(D)$;
- the witness term $1/((20+2\sqrt{5})(D+1))$.

The witness term never binds. The cap θ_0 is independent of D and no strengthening of Lemma W removes it. By Example 2.2, the constant $1/(D+1)$ cannot be improved using only "Case-D BFT triple + range $\leq D$ ". Whether it improves under the global hypothesis $\delta(P) \leq C_{\text{BFT}} + \theta$ is open. That hypothesis is vacuous for $D \leq 3471$ by the theorem itself. If Lemma 4.2's $M^2/441$ were used in place of Proposition 5.3, the D -uniform range would end at $D = 17$.

Remark 6.2 (where finiteness enters). The bound cannot hold for infinite posets. [BFT95, §1 (the poset Q) and Thm. 1.4] give thin infinite posets of range 2 (for example the Fibonacci poset, $x_i \prec x_j$ iff $j - i \geq 2$) whose balance constant is exactly C_{BFT} . Let F_m be its finite truncation to $x_1, \dots, x_m$. At the central triples of F_m we have $N(x, y) = \emptyset$ and the bridged $2+1$ structure of Lemma 4.1 holds. Heuristically, therefore, Lemma W and Lemma 4.1 are not where the finite and infinite cases differ; we have not made the passage to infinite posets (whose linear extensions are defined only as limits) rigorous. In the proof above, finiteness is used only in the witness step. Lemma 5.1 needs the boundary relations $d_1 \geq 0$, $d_1 + d_2 \geq 0$, $d_{n-1} + d_n \leq 0$, and Proposition 5.2 needs a bottom element v_1 with $d_1 \geq 1/(D+1)$. In F_m the displacement M is attained at the ends and stays bounded away from 0, while every fixed central displacement tends to 0.

Corollary 6.3 (effect on [KSFT, Thm. 1.2]). [KSFT, p. 5] says only that Theorem 1.2 ($\delta(P) \geq C_{\text{BFT}} + \epsilon$ for all finite non-chains) follows from combining Thms. 1.3, 1.4 and 1.5. The natural combination, which is our reconstruction and is not written out in [KSFT], is this. Thm. 1.4 handles width greater than K . Thm. 1.5, which [KSFT] imports from the preprint [AK25b], needs width $w < K'$, so it is applied with $K' = K + 1$; it handles range greater than $L = L(K + 1, \cdot)$. Thm. 1.3 is then applied at $D = L$. Using Theorem 1.3 in place of Thm. 1.3 in this combination gives $\epsilon = \min(\theta_0, \theta_W(L))$, which is about $0.0236/(L+1)$ once $L > 3471$. The factor $(L+1)^{-4(L+1)}$ becomes $\asymp 1/L$, and the size of ϵ is governed by L alone. The

paper does not print L [KSBFT, p. 5], and it omits the derivation of its (1.5), so we give no numerical value. The conditionality of Thm. 1.4 on [AK25a] and Haqi's preprint [Haq26], and of Thm. 1.5 on [AK25b], is unchanged.

7. ON FACT I3: APPENDIX A OF [KSBFT] AND THE VALUE 67/242

We checked the proof of [KSBFT, Lemma 2.3] (Appendix A) against the preprint of [BFT95]. It imports the following results:

- Fact I1;
- [BFT95, Lemma 2.1], the Kahn–Saks inequalities (2.1)–(2.6) for the sequences a_i, b_i , including the log-concavity (2.6), which rests on the Alexandrov–Fenchel inequalities [KS84]; these are the hypotheses of the packing lemma;
- [BFT95, Lemma 2.2], the packing lemma, which gives $h(y) - h(x) \geq H(p, t)$ for the packed height function H ;
- [BFT95, Lemma 2.3], the monotonicity lemma: $H(B, \epsilon)$ is decreasing in ϵ for $B \leq 1/3$, except in case (ii) with $k = 1$ and $\epsilon \leq 1/\sqrt{2}$, which occurs only when $B \geq 1 - 1/\sqrt{2} \approx 0.293$;
- [BFT95, Lemma 6.1]: if $x \parallel z$ and $h(z) \leq h(x) + 2$, then $\mathbb{P}[f(z) < f(x)] \geq 3/22$;
- [BFT95, Lemma 6.2]: if x, y, z are pairwise incomparable and $h(z) \leq h(x) + 2$, then $\mathbb{P}[f(x) - f(y) \geq 2] + \mathbb{P}[f(y) - f(z) \geq 2] \geq 1/11$. Its proof uses Lemma 6.1 and needs no bound on p or p' ;
- the standard fact that $t > 0$, i.e. some linear extension has y immediately before x (list the down-set $I = \{u : u \prec x\} \cup \{u : u \preceq y\}$ first, ending with y , which is maximal in I because $x \parallel y$; then x , all of whose predecessors lie in I ; then the rest), which the appendix uses when it takes $t \in (0, 1]$.

Everything else in the appendix is algebra, which we re-derived by hand and checked numerically:

- the closed forms (A.2) for H , which agree with the definition of packed sequences to $3 \cdot 10^{-13}$ on a grid;
- Claim A.1;
- the boundary identity and bound (A.5);
- both sub-cases of Claim A.2, including the packed-case conditions $1 + 2t + 2t^2 \leq 1/p \leq (1+t)(1+2t)$;
- the final inequality, $5 - 11 \cdot 0.2764 + \frac{1}{22} = 2.00505 > 2$.

One sentence of the appendix is ambiguous. It reads “the bounds on q imply $3/5 < v_q < 1$ ”. If “the bounds on q ” means $q \in (1/5, 3/10)$, the implication fails near $q = 3/10$ ($v_{3/10} \approx 0.565$). On the range $q \in [6/11 - 0.2764, 0.2764]$, which the appendix has also just established, it holds: there $v_q \in [0.618, 0.636]$. So the conclusion is correct.

The value 67/242. The last line of the appendix reads $5 - \frac{11}{2}(p + p') + \frac{1}{22} > 2$, and it holds for every threshold $T < 67/242$. Every other step (Claim A.1's use of $T < 1 - 1/\sqrt{2}$, the range of v_q , Claim A.2's $32/9 < 1/T$, and the first branch, which needs $T < 37/132$) remains valid at $T = 67/242$. So the argument proves $\max(\delta(P; x, y), \delta(P; y, z)) \geq 67/242 \approx 0.276860$, with a non-strict inequality: at $T = 67/242$ the chain gives exactly $h(z) - h(x) \geq 2$. The appendix's claim that it “yields the stronger lower bound 67/242” is therefore correct, modulo the same imports. Note that this improvement rests on the *proof* in Appendix A, not on the statement of [KSBFT, Lemma 2.3]. It therefore imports, beyond Facts I1–I4, everything listed above: [BFT95, Lemmas 2.1, 2.2, 2.3, 6.1, 6.2] and $t > 0$. With it, Lemma 4.1(i) holds for every $\theta < \theta'_0 := 67/242 - C_{\text{BFT}} = 4.663 \cdot 10^{-4}$. Lemma 3.1 and Lemma 3.2(c) hold at threshold $67/242 < e^{-1}$ verbatim. Theorem 1.3 then improves to

$$\delta(P) \geq C_{\text{BFT}} + \min(\theta'_0, \theta_W(D)),$$

which is $\geq 67/242$ for $D \leq 49$.

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